Weekly Site Report

BLACK ROCK WEEKLY REPORT

Weekly PSD Report
EPIROC SA

Contents

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Weekly Service Report – Black Rock Mine Operations

Report type:	**Weekly**	**Submitted to:**	**Sello Taku and Sipho**
Week Date:	20 February 2026	**End Date**	26 February 2026
Reported By:	Philip Moller		
BEV:			
Service Tech:	Ivan Phiri	**Service Tech:**	Mathews Mosemane
Service Tech:	Patricia Bulelwa being replaced by Iphriam Mokofane	**Service Tech Assistant:**	Patrick Yoko (Batteries)
Specialist:	Frikkie van der Walt (Batteries)		
Field Service:			
Service Tech:	Lucas Slaffa	**Service Tech:**	Phineas Sibanda
CAS Project:			
Team Leader:	Wisani Shilenge	**Service Tech:**	Phineas Sibanda
Service Tech:	Julius Manamela	**Service Tech:**	Joseph Mampa
Certiq Project:			
Team Leader:	Segunda Fernado	**Service Tech:**	
Mobilaris:			
Team Leader:	Rupert Grobler		

Report Summary

General Summary

The purpose of this report is to give an overview of all the Epiroc activities at Blackrock Mine Operations (BRMO).

BEV - Critical Breakdown Summary

- DT171 - A-frame that was previously damaged failed again.
- DT-147 Isolation resistance fault WM801XV2 5590025928 cable down to ground.

- Battery breakdowns increased. Currently busy with TMS hyper care. Most of the breakdowns are TMS related. We are having coolant pump failures. New pumps ordered and will arrive 07 March. Changing coolant and coolant hoses on services.
- I looked at the job cards on click view to see if I can get more info and do root cause analysis on the machines and look for re-occurring faults or trends. Unfortunately, it seems like job cards are not being filled in properly anymore. We are left with limited reporting to do this kind of RCAs.

CAS L9 Implementation - Critical Issues Summary

- Lab scale tests with new software currently underway.
- Invoicing and sign off to be completed.
- MH4.0 software to be loaded on the BEV machines. Currently busy with the lab scale tests and approval of the software.

Certiq & Mobilaris - Critical Issues Summary

Passport 360 Compliance

- BEV Technicians: **94%**
- CASL9 (Gloria): **95%**
- CASL9 (Nch2): **93%**
- CAS L9 (Nch3): **93%**
- Certiq & Mobilaris N2: **87%**
- Certiq & Mobilaris Gloria: **87%**

BEV Report

Weekly machine utilization summary:

Friday 20 February 2026:

1. FL 98 -
2. FL 99 -
3. FL 107 -
4. FL 108 -
5. FL 112 -
6. FL 113 –Morning shift: Breakdown Mechanical – No feedback (1H15M)
7. DT 146 – Morning shift: Auto Electrical System – 24v – No feedback (54M)
8. DT 147 - Afternoon shift: Breakdown Mechanical – Remove front tyres to repair hydraulic oil leak (3D22H10M)
9. DT 149 –
10. DT 150 -
11. DT 162 -
12. DT 163 - Morning shift: Auto Electrical System – 24v – No feedback (1D2H24M)
13. DT 172 -

Saturday 21 February 2026:

1. FL 98 -
2. FL 99 -
3. FL 107 -
4. FL 108 -
5. FL 112 -
6. FL 113 -
7. DT 146 -
8. DT 147 -Afternoon shift: Breakdown Mechanical – Remove front tyres to repair hydraulic oil leak (3D22H10M)
9. DT 149 –
10. DT 150 -
11. DT 162 -
12. DT 163 - Morning shift: Auto Electrical System – 24v – No feedback (1D2H24M)
13. DT 172 -

Sunday 22 February 2026:

1. FL 98 -
2. FL 99 -
3. FL 107 -
4. FL 108 -
5. FL 112 -
6. FL 113 -
7. DT 146 -
8. DT 147 -Afternoon shift: Breakdown Mechanical – Remove front tyres to repair hydraulic oil leak (3D22H10M)
9. DT 149 –
10. DT 150 -
11. DT 162 -
12. DT 163 -
13. DT 172 -

Monday 23 February 2026:

1. FL 98 - Morning shift: Breakdown Mechanical – No feedback (6H11M)
   - Afternoon shift: Breakdown Electrical – TCU error. Replace TCU fuse (20H38M)
2. FL 99 -
3. FL 107 -
4. FL 108 -
5. FL 112 -
6. FL 113 -
7. DT 146 -
8. DT 147 -Afternoon shift: Breakdown Mechanical – Remove front tyres to repair hydraulic oil leak (3D22H10M)
9. DT 149 –

10. DT 150 - Afternoon shift: Breakdown Electrical – No feedback (40M)
11. DT 162 - Afternoon shift: 3rd Party Supplier Fault – STRATA (2H)
12. DT 163 -
13. DT 172 -

Tuesday 24 February 2026:

1. FL 98 -
2. FL 99 -
3. FL 107 -
4. FL 108 -
5. FL 112 -
6. FL 113 - Night shift: 3rd Party Supplier Fault – STRATA (3H8M)
7. DT 146 -
8. DT 147 -Afternoon shift: Breakdown Mechanical – Remove front tyres to repair hydraulic oil leak (3D22H10M)
9. DT 149 – Night shift: Breakdown Electrical – No feedback (2H39M)
10. DT 150 - Night shift: 3rd Party Supplier Fault – STRATA (3H3M)
11. DT 162 -
12. DT 163 –Night shift: Breakdown Tyre Bay: Chassis: Tyres – Flat tyre (6H44M)
13. DT 172 -

Wednesday 25 February 2026:

1. FL 98 -
2. FL 99 -
3. FL 107 -
4. FL 108 -
5. FL 112 -
6. FL 113 – Morning shift: 3rd Party Supplier Fault – STRATA (3H44M)
7. DT 146 -
8. DT 147 -
9. DT 149 –
10. DT 150 -
11. DT 162 -
12. DT 163 - Morning shift: Breakdown Tyre Bay – Tyre replacement (14H10M)
13. DT 172 -

Thursday 26 February 2026:

1. FL 98 - Morning shift: Breakdown Mechanical – No feedback (52M)
2. FL 99 -
3. FL 107 -
4. FL 108 - Morning shift: Breakdown Mechanical – No feedback (11H44M)
5. FL 112 -
6. FL 113 - Morning shift: Breakdown Boilermaker – Replace half arrows (1H30M)
   - Night shift: Breakdown Mechanical – No feedback (47M)
7. DT 146 -

8. DT 147 - Afternoon shift: Breakdown Electrical Ongoing - HV cable test to ground. Need to order new cable. (20H14M)
9. DT 149 –
10. DT 150 -
11. DT 162 -
12. DT 163 – Afternoon shift: Breakdown Mechanical – No feedback (23M)
13. DT 172 – Morning shift: Breakdown Electrical: Auto Electrical System – 24v – No feedback (1H17M)

Planned work for the coming week:

NCH3:

- Charger 3 monthly services
- Battery monthly services.

NCH2:

- Fault finding on FL0122, machine CAN fault after dummy cab was installed, busy with fault finding. Neet to commission machine.
- Commission RT0061 (New close cab machine)

Gloria

- Safety systems audit to be done. 2 Machines left.
- Audit of RT0045 and RT0046. Machines have low availability.

Improvement and actions:

1. We need to look at the data of the charger interrupts. This function was tasked with Desmond. We will start seeing data coming through this week.

2. Post 3,6 and 7 cables replaced, Piet ordered the rest of the cables.

3. Auxiliary motor spline grease TSNB not done. Only DT147 completed. We need to schedule the machines to complete this campaign. Grease already on site. Piet and myself to schedule machines. We started the discussion.
4. Regeneration knob set to 100% completed. Need to sign off change management and attendance register for operators and artisans underground.
5. Need to physically check status of chargers and compare to report. Chargers checked. All modules put back except for charger 2 with a faulty module.
6. VCA, VCB cables and discharge resistors identified machines to be replaced. Parts ordered. - Wimpie and Piet, can you please update me on what machines resistors and connectors was replaced.

   DT147
   - Discharge resistor not working
   - VCA connector worn
   - VCB HVIL pins pushed back
   - VCB pins pushed back
   - VCB connector clamps broken

DT149

- Discharge resistor not working
- VCA connector worn

DT 150

- Main switch need to be replaced
- VCB insulator missing
- VCB pins have Arc marks
- VCA connector worn
- VCA cable trunking loose

DT171

- Discharge resistor not working
- VCB connector insulator missing

FL107

- Discharge resistor not working
- VCA plug worn
- VCB insulator missing
- VCB connector clamps broken

FL98

- Discharge resistor not working
- VCA connector worn
- VCB insulators missing

FL108

- VCA connectors joint inside plug, need to be replaced
- VCA plug worn
- Insulators missing

7. Audits to be done on all A frame retaining bolts of dump trucks. Could not do this. Currently busy with Brake wear measurements on BEV machines.

DT147

Right rear – 3.28mm
Left rear – 3.18mm
Front Right – 3.4mm
Front left – 3.12mm

DT172

Right rear – 4.72mm
Left rear – 4.72mm
Front Right – 4.76mm
Front left – 4.7mm

DT150

Right rear – 4.0mm
Left rear – 3.3mm
Front Right – 2.9mm
Front left – 2.8mm

DT163
Right rear – 2.9mm
Left rear – 2.8mm
Front Right – 2.5mm
Front left – 2.1mm

DT149
Right rear – 3.6mm
Left rear – 3.7mm
Front Right – 3.3mm
Front left – 2.7mm

FI99
Right rear – 1.6mm
Left rear – 1.7mm
Front Right – 1.3mm
Front left – 1.3mm

TSNB completion percentile

B4, B5 and chargers

B4 Campaign	
Campaign	Percentile
DCDC	95%
MH 6.3	80%
TMS SW V2.0.3.0	20%

Charger SW Campaign	
SW V3.4.7	90%

B5 Campaign	
Campaign	Percentile
DCDC	93.33 %
MH 6.3	53.846%
TMS V2.0.3.0	0%

8.
9. Busy with TMS hyper care on all batteries.

Services

Nchwaning 3

Maintenance Schedule Mar 2026				
Friday	Monday	Tuesday	Wednesday	Thursday
27-Feb-26	2-Mar-26	3-Mar-26	4-Mar-26	5-Mar-26
	DT0160	DT0147	DT0154	
	FL0121			
HD0056	HD0057	HD0058	HD0061	HD0062
RT0048	RT0049	RT0051	RT0052	RT0054
SR0026	SR0028	SR0034	SR0035	SR0038
	UV0126	UV0131	UV0134	UV0135
	UV0107	UV0112	UV0122	UV0132
	LD0445	LD0512	LD0520	LD0555
	VPY-00011 (BY0010)	VPX-00010(BY0004)	VPY-00031 (BY0012)	VPX-00015(BY0031)
Highlighted in Blue Involve STRATA Sats				

Battery and Charger Status Report

General summary

ST14 – B4 battery packs

- We currently have 10 x ST14 – B4 battery packs and 6 x ST14 machines operational underground. This is equal to the committed ratio of 1.6 batteries per machine.

MT42 – B5 battery packs

- We currently have 12 x MT42 – B5 battery packs operational underground. This is above the committed ratio of 1.6 batteries per machine.

160 kW Chargers

General

- DCDC campaign 95% complete. Initial indication from team is that charger stops significantly reduced.

- TSNB software V6.3 to be installed on batteries. (level 3 safe state)
- Fleet number stickers ordered and installed on batteries.

Table 1: Battery pack availability for week 2025

Battery Type	Battery ID	Status	Comment
B4 - ST14	VPY-00011	Working	27 Feb Condenser clocked. Clean condenser with blower (1HR)
B4 - ST14	VPY-00031	Working	
B4 - ST14	VPY-00051	Working	25 Feb Coolant pump failed swap TMS with VPY-00076 (1D)
B4 - ST14	VPY-00048	Working	
B4 - ST14	VPY-00049	Working	
B4 - ST14	VPY-00088	Working	26 Feb TMS cooling pump fuse replaced, Loose connection on plug. (4HRS)
B4 - ST14	VPY-00086	Working	
B4 - ST14	VPY-00076	Breakdown	Battery will be replaced by VPY-00034
B4 - ST14	VPY-00083	Working	
B4 - ST14	VPY-00041	Working	
B5 - MT42	VPX-00016	Breakdown	VPC#3 is in a safe state. Waiting for Scania.
B5 - MT42	VPX-00015	Working	
B5 - MT42	VPX-00023	Working	
B5 - MT42	VPX-00017	Working	
B5 - MT42	VPX-00010	Working	
B5 - MT42	VPX-00050	Working	23 Feb Replace coolant hose and re pressure system (3HRS)
B5 - MT42	VPX-00031	Working	
B5 - MT42	VPX-00036	Working	
B5 - MT42	VPX-00028	Working	
B5 - MT42	VPX-00026	Working	
B5 - MT42	VPX-00024	On DT-171	Battery out of production. On DT-171 (breakdown MT42)
B5 - MT42	VPX-00048	Working	
B5 - MT42	VPX-00044	Working	

Battery charger summary:

Charger 1

Module 2 switch off and pulled out. All fans working on the 3 modules that is in used. Trip over current in mini sub if it is switch on.

Charger 2

Module 2 switch off and pulled out. All fans working on the 3 modules that is in used. Module 2 faulty.

<u>Charger 3</u>
Module 4 switch off and pulled out. All fans working on the 3 modules that is in used. Trip over current in mini sub if it is switch on.

<u>Charger 7</u>
Module 4 switch off but still in place. All fans working on the 3 modules that is in used. Switched on.

<u>Charger 4</u>
Module 1switch off and pulled out. All fans working on the 3 modules that is in used. Trip over current in mini sub if it is switch on.

<u>Charger 5</u>
Module 4 switch off and pulled out. All fans working on the 3 modules that is in used. Trip over current in mini sub if it is switch on.

<u>Charger 6</u>
Module 1 switch off and pulled out. All fans working on the 3 modules that is in used. Trip over current in mini sub if it is switch on.

<u>Charger 8</u>
Module 4 switch off and pulled out. All fans working on the 3 modules that is in used. Trip over current in mini sub if it is switch on.

Critical Issues on Batteries

- VPY-00011 started to show signs of overheating. Condenser was cleaned and the battery was cooling down fine.
- VPY-00051 was overheating. The cooling pump failed. The TMS was swapped with VPY-00076 (The battery that will be replaced by VPY-00034)
- VPY-00088 Cooling pump fuse trip. Repaired loose connection on pump connector.

FST Report Summary

CAS Report Summary

Report Type:	Weekly	**Project Leader:**	Hennie van Niekerk
Start Date:		**Report By:**	
End Date:		**Contract No:**	BPE145D

Progress Update